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Designing robust scale-free networks under targeted link attack using local information

Marco Tomassini

Information Systems Institute, University of Lausanne, Switzerland

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ABSTRACT

We study the effects of deliberate attacks to important links in scale-free networks and propose simple strategies to mitigate the damage. While strategies that require global network information and heavy computation have been proposed, here we focus on heuristics that make use of local information only and are much easier to compute. Using model scale-free networks and under the constraint of an invariant degree distribution, we show by numerical simulation that our approach in the average allows to enhance the network robustness notably, as measured by the integrity of the largest connected component. Because it is fast, the approach is also applicable to real-world networks of which we give two typical examples. Overall, the results are equivalent to those obtained in more complex settings that require global network knowledge and are much more computer-intensive, an important point for large networks.

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1. Introduction

Extended networks of all kinds pervade our society, from infrastructures such as transportation and communication, to social and economical relationships. Even the biological functioning in living cells is largely dictated by the networked interactions between fundamental constituents such as proteins and nucleic acids. A corpus of disciplines called Network Science has emerged in the last two decades to make use of the large amount of data readily available to model, describe, and analyze such complex networks [1]. The vulnerability of complex networks when submitted to random failures or intentional attacks is a particularly important problem because damage inflicted to such structures could impair their functioning to the point of rendering the network ineffective. Although resilience has long been studied in engineering especially through the use of redundancy and replication, already in 1970 Frank and Frisch [2] mentioned some of the conditions that must hold for a network to remain operational after un attack. In the context of model complex networks the issue was first raised by Albert et al. in a pioneering study [3]. They showed that scale-free networks, which approximately describe the structure of many real networks, are resilient against attacks or failures that target random vertices but they quickly become disconnected and disintegrate if attacks are targeted at the highest degree nodes. Following this important discovery, several techniques have been suggested to mitigate the effects of attacks on complex networks by enhancing network robustness and the field has been an active one to date. More recent work, including biological networks which are not dealt with here, can be found in the reviews by Gosak et al. [4] and by Jusup et al. [5]. Besides, a new trend is emerging which is not treated here either, i.e., extending vulnerability analyses to multilayer and interdependent networks. Recent work in this field can be found, for example, in [6–8].

Here we mainly follow the line of reasoning of Ref. [9] in which it was argued that robustness against attacks targeted at the most connected nodes could be significantly increased by rewiring a certain amount of links, leaving each node's

E-mail address: marco.tomassini@unil.ch.

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degree constant and thus without changing the degree distribution. The latter constraint is reasonable in two ways: changing a node's degree for an existing network is generally more costly than just rewiring a link and link swaps can be kept local in order to be more technically feasible. In addition, if the methodology is used to design a new network from scratch for a given degree distribution, then optimizing the link rewiring may lead to designs that possess an adequate robustness to high-degree node attacks. The results presented in [9,10] show that the methodology indeed leads to more robust network structures, either real or simulated, against attacks directed at the most connected nodes. Several other ways of rendering networks more robust against malicious attacks have been suggested, e.g., [11–15]. They differ from [9] either in the measure of robustness used, or in the heuristic suggested for link rewiring, or both. We shall discuss work relevant to the present study in the following where the terms node or vertex and link or edge are to be considered synonyms.

While the focus of a large part of the previously described work was on attacks to the network nodes, attacks targeted at the network links have been comparatively less studied although links are equally important for the integrity of the network. For example, to inflict damage to a power line it might be simpler to cut cables between two electric pylons or to destroy a pylon rather than targeting a power plant. Also, in a war situation, a given airline route could be suppressed because of possible missile ground attacks. The destruction of communication cables in communication networks is another example. Work that is relevant for our purposes here is described next. Early in 2002 Holme et al. [16] made a quite extensive analysis of vertex and link attack vulnerability in complex networks based on several criteria for choosing the sequence of nodes and edges to be suppressed such as high vertex or link degree, high vertex or link betweenness, or high edge degree product. However, they do not describe techniques to try to enhance robustness in the paper. In a more recent study [17], Zeng and Liu, extending the methodology of Schneider et al. [9] to network edges instead of vertices, simulated attacks to high betweenness edges of Barabási–Albert (BA) scale-free networks [18]. In another recent paper, Liu et al. [15] compare several network robustness measures based on connectivity, giant component size, and graph matrix spectra using different classes of attacks to nodes or links including degree, betweenness, and closeness centrality. For another type of approach, Van Mieghem et al. [12] and Pizzuti et al. [19] make use of spectral graph theory in the study of link attack on the robustness of a network. In particular, [19] compares link attacks based on spectral decomposition with structural attacks like those described in, e.g., [15,16].

Most of the works discussed above make use of attack types or robustness measures that require a high computational effort and global knowledge of the network structure. As these conditions may constitute hard constraints on what can be done, in this work we investigate what it is possible to achieve in terms of network resilience to targeted link attacks using easy to obtain local information and a robustness measure that is also fast to compute. We find that it is indeed possible to obtain significant results under this scenario provided that the original networks are rather sparse.

The structure of the article is as follows. In the next section we review the main attack types to nodes and links and describe our choices. Next the edge swap heuristic and its local variant are presented, together with the results obtained. This includes a comparison with another robustness metric and an analysis of the structure of the robustness-enhanced networks. We then study the effectiveness of the approach on two real-world networks. Finally, we summarize our findings in the conclusions section.

2. Targeted attacks types and network robustness

The problem we want to tackle is the numerical simulation of the robustness of a given network against malicious attacks targeting the edges of the network. This requires a choice for the network model, the type of attack, and the robustness measure to be used to evaluate the overall resilience of the network.

The model network will be of the Barabási–Albert type with degree distribution $P(k) \sim k^{\gamma}$, $\gamma \approx 3$. Although the exponent γ is fixed and does not cover the full range observed in real cases, they are easy to generate and sufficiently similar to many real networks to be representative of the main features of interest. Moreover, they have been adopted in several previous studies, e.g., [10,13,15,17] and thus allow for comparison. Alternatively, one could use random networks of the scale-free type with a given exponent by generating networks from a suitable degree sequence through the configuration model construction [20]. Plain random networks of the Erdos–Renyi type are less interesting in our context because their homogeneous degree distribution makes them less vulnerable to targeted attacks; besides, this type of network is not a good model for the vast majority of actual networks. For the simulations we use networks of size $N = 200$ nodes. This makes the lengthy computations affordable while maintaining the significance of the results, as shown for example in [21], where it was found that there are almost no differences for $N = 100, 500, 1000$.

Concerning the intentional attack when it is to be directed at the network's nodes, two choices are often suggested: either target the nodes in descending degree order, i.e. targeting hubs first, or attacking nodes in descending vertex centrality order. The two are rather different in spirit. Vertex centrality is not adequately represented by vertex degree because this is a purely local property that ignores the structure of the rest of the network. Still, nodes with many connections are an interesting target for a malicious attack. The advantage is that node degree is easy to compute and usually easy to infer when attacking a partially unknown real network. More complex vertex centrality measures, such as betweenness or closeness, are better measures of centrality and thus of node importance in the whole network. However, they are computationally much more expensive; betweenness, for instance, requires in the worst case a number of operations $O(N^3)$ for a network with N nodes. The other global centralities measures such as closeness or eigenvector

centrality require a similar amount of computational effort being based either on all shortest paths pairs (closeness) or on the eigenvectors and eigenvalues of the adjacency matrix or the laplacian, the latter requiring matrix inversion and thus $O(N^3)$ operations for dense graphs. Besides the high computational effort, there is another reason why a centrality measure such as betweenness might be less than adequate to identify important nodes to attack. Indeed, to compute betweenness one must have global knowledge of the network. While this is obviously granted for model networks generated with an algorithm, it becomes difficult for real networks, where parts of the network might be unknown.

The logic is the same when targeting edges instead of nodes. The analogous of node degree $d(i)$ of node i is the *edge degree* of edge (a, b) usually defined as $d(a) + d(b)$ or $d(a) + d(b) - 2$ to account for the fact that a and b are neighbors. One could also use the *degree product* $= d(a) \times d(b)$, as well as a few other choices, all first suggested in [16]. The same is true for centrality, using the edge-based version of betweenness for instance. In the following, for the reasons given above, we shall use the edge degree to rank edges for malicious attacks such that the edge having the highest degree is attacked first. It is fair to remark that sorting edges according to their degree also requires global network knowledge as for betweenness or closeness. However, in practice, identifying high degree edges to attack, not necessarily in ranked order, is usually easy in real networks and the effects will be close to those produced by the strictly ranked order.

There are several ways to compute network robustness during and after attacks. Ref. [15] contains an interesting comparison of most of them. In general, robustness measures can be based on percolation and connectivity ideas or on spectral graph theory. Among the first type the R index and the *efficiency* are often used. The R measure has been proposed in [9]. It is defined as the size of the largest connected component after a series of attacks that remove the nodes in descending degree order:

$$R = \frac{1}{N} \sum_{Q=1}^{N} S\left(\frac{Q}{N}\right), \tag{1}$$

where N is the number of nodes in the network, Q is the number of nodes removed, and $S(Q/N)$ is the size of the largest connected component after removing a fraction Q/N of the nodes. After having removed a fraction of nodes, the degrees should be updated for the next attack. This definition has been extended in [17] to attacks on links. In this case N is to be interpreted as the number of links of the graph and Q is the number of removed links instead of nodes. Here too, R is recomputed after each link removal. A higher value of R indicates a more robust network.

Efficiency, another frequently used robustness measure proposed in [22], is defined as follows:

$$E(G) = \frac{1}{N(N-1)} \sum_{i \neq j} d_{ij}^{-1}, \tag{2}$$

where G is a graph with N vertices and d_{ij} is the shortest path between vertices i and j. If there is no connection between a pair i, j then $d_{ij} = \infty$ and the term contributes nothing to the efficiency. Clearly efficiency is harder to compute than R for it requires to calculate all the shortest paths and it must be recomputed after each attack. In some works the *integral* version of efficiency defined in [13] as $IntE = \frac{1}{N} \sum_{Q=1}^{N} E\left(\frac{Q}{N}\right)$, where N is either the number of vertices or the number of edges depending on whether we attack vertices or edges respectively, has also been used. The higher E or $IntE$, the more robust the network. It can be remarked that while R is related to the topological integrity of the networks, efficiency measures the network's communication capabilities. The two are related but are not the same, as we will see.

Finally, it exists a third class of measures based on the spectral decomposition of the adjacency matrix or of the graph laplacian, i.e., on finding the eigenvectors and eigenvalues of these matrices. These approaches are interesting and have been used in several works, e.g., [12,14,19,23] but, because they are rather expensive computationally and for lack of space, we will not use them here.

3. Degree-preserving edge swap for mitigating edge attacks

Several approaches to enhanced robustness of networks against targeted attacks make use of degree-preserving randomization [9,10,13–15,17]. This consists in randomly selecting two links and swap them if they do not lead to multiple links or have a node in common. This operation lets the degree of each node and the full degree distribution unchanged. The choice is motivated by the fact that, in many cases, especially for existing networks, it is more advantageous to preserve the node degree, and thus the degree distribution as increasing the number of edges or changing the degree distribution could be technically infeasible because of hard constraint on costs and capacity. For instance, rerouting a flight or rerouting messages would arguably be easier than adding a new route or a new transmission line. A schematic view of the swap operation is given in Fig. 1 and the corresponding logic is that of pseudocode 1.

The number of iterations is chosen by trial and error such that the R index improvement becomes less than a given small $\epsilon > 0$ after that number. Clearly, such a hill-climbing heuristic cannot guarantee global optimality and so when we speak of "optimized networks" in the sequel it must be understood as a satisfactory local optimum. The time complexity of algorithm 1 is dominated by the edge sorting operation needed for the computation of R which is $O(MlogM)$, M being the number of edges of the graph. If edge betweenness is used instead of edge degree as the target criterion then time complexity goes up to $O(N^2)$ in the most favorable case of very sparse graphs ($M \approx N$). More complex versions of edge rewiring have been described in [14] and Monte Carlo approaches have been suggested in [13,24]. However, after some experimentation we have found that the simple swap described here and first used in [9] works sufficiently well for the task thus avoiding having to set a number of new parameters.

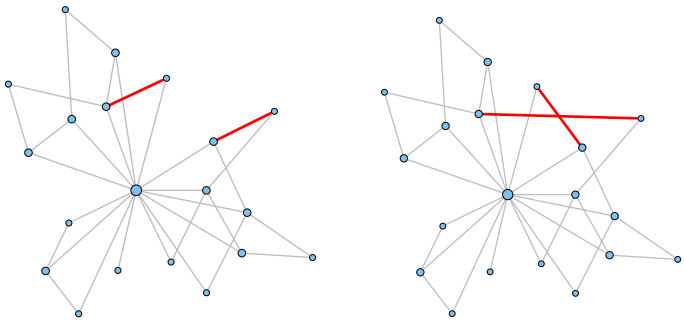

Fig. 1. The edge swap operation. Two edges are chosen at random in the original graph (thick lines, left image) and they are swapped as indicated in the right image.

Algorithm 1 Edge Swap Heuristic

```
Create graph G
Compute R(G)
for i = 1 → number of iterations do
    Choose edges e_ij, e_lk ∈ G uniformly at random
    Swap edges e_ij, e_lk obtaining a new graph H
    Compute R(H)
    if R(H) > R(G) then
        G ← H
    end if
end for
```

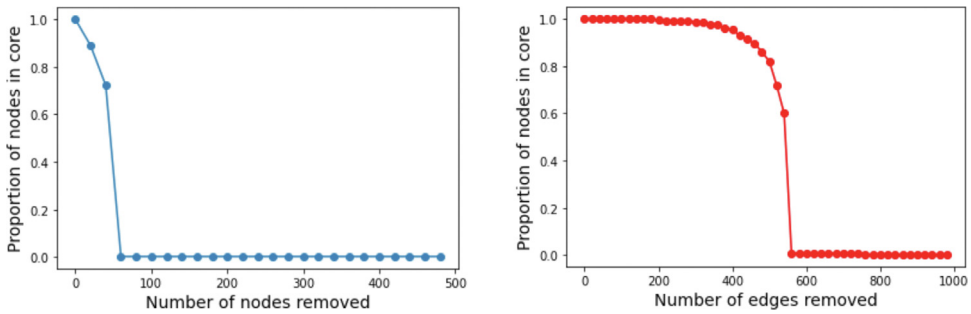

Fig. 2. Attack on the higher degree nodes (left) and higher edge degree edges (right) of a BA graph with N = 500 nodes and $\bar{k} \approx 4$. The size of the largest connected components is given on the y-axis as a function of the number of nodes or edges removed. It can be seen in the simulation that attacking high degree vertices is more effective than attacking high degree links in dismantling the same network.

3.1. Robustness of BA networks to edge attacks

Attacks to network edges are a little different from attacks directed to vertices for all network types, not only BA graphs, which we use here. In fact, each time a node is removed all of its links are removed as well while the removal of a link does not affect the other links and the number of vertices remains constant. So, even though the network falls apart in both cases after a sufficient number of removals, the same network is intrinsically more robust when attacks are targeted at links rather than vertices. This can be seen in Fig. 2 where we recall that nodes and links are attacked in descending degree and edge degree order respectively. As already indicated, the sorted list of vertices or edges to attack is updated after each removal since the structure of the network changes.

Another obvious effect that is nevertheless worth pointing out is that, in general, the higher the mean degree of the network, the higher the network robustness against link attacks. This is shown is Fig. 3 for BA graphs with 500 nodes and mean degree $\bar{k} \approx 4, 6, 8$. It is clearly seen that the fraction of edges that need to be destroyed before the graph falls apart increases with the mean degree of the graph. Thus a trivial way of making a network resistant to attacks targeted at the most connected network's edges is simply to increase its mean degree. One could call this "intrinsic robustness" but, as it is only based on redundancy, it cannot be used in practice because of high cost and technology constraints. Besides, most real-world networks are sparse, i.e., they have a low mean degree, especially technological networks [20].

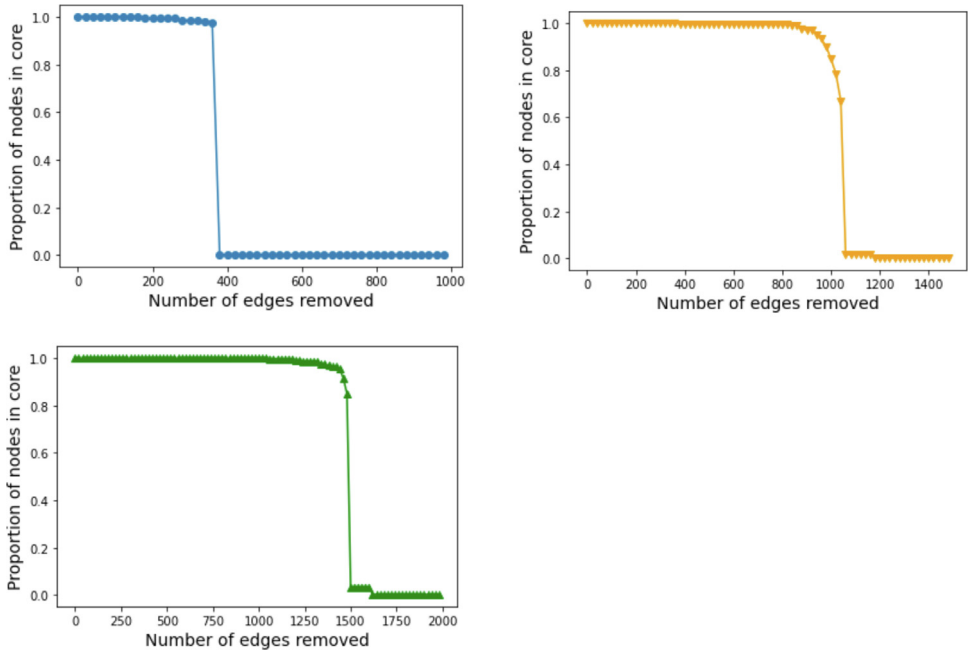

Fig. 3. Size of the largest connected component as a function of the number of edges removed for BA graphs of size N = 500 and mean degree $\bar{k} \approx 4$ (left), 6 (middle), and 8 (right). The attack is on high edge degree (note the different horizontal scales). For the same network size, it appears that one needs to remove more edges as the mean degree increases before the network falls apart.

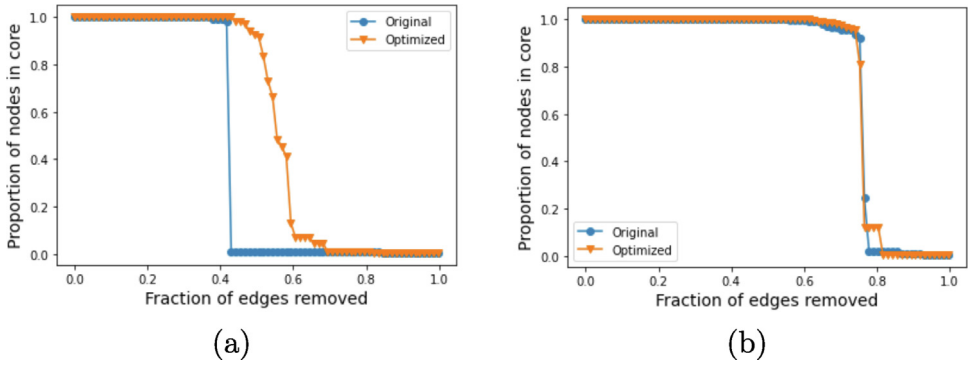

Fig. 4. Size of the largest connected component as a function of the fraction of edges removed for BA graphs of size N = 200 and mean degree $\bar{k} \approx 4$ (a) and $\bar{k} \approx 8$ (b). The attack is on high edge degree. Here, as in the previous Fig. 3, the simulations show that edge rewiring provides protection against link attacks when the mean degree is low, i.e., the graph is sparse and vulnerability is high. At higher mean degrees the network is more difficult to dismantle and rewiring does not help to further increase robustness.

In [17] Zeng and Liu extend the R robustness measure to link attacks (see Section 2) in order of decreasing link betweenness, an attack type first proposed by Holme et al. in [16]. However, for the reasons explained in Section 2 here we shall follow a simpler strategy not considered in [17] which consists in attacking edges in decreasing edge degree value. In fact, to compute edge betweenness knowledge of the whole network is needed, something that might not be possible in practice for real networks. Besides, the computation is very demanding. On the other hand, single link features are both easier to observe and require a very moderate amount of computation, although the drawback is that these local features are not necessarily representative of the global network structure. We note that, strictly speaking, we must sort the edges in decreasing edge degree order, which also requires global network knowledge; however, in a real network situation a similar attack might be performed simply by targeting the edges that appear to be the most connected to an external observer, an approximate information that is often easily available.

Fig. 4(a) depicts the typical result of using the edge swapping technique of algorithm 1 on BA networks of size 200 nodes and $\bar{k} \approx 4$ when targeting the edges of highest degree. The number of iterations of the swapping heuristic was 10^4 in all simulations. The overall mitigation of the damage inflicted by targeted edge attacks corresponds to the area between the optimized and the original curves. The mean improvement of the robustness index over twenty runs of the

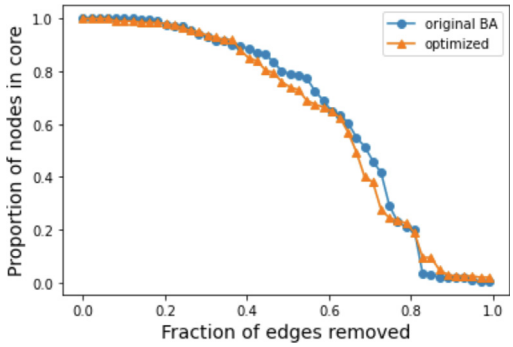

Fig. 5. Fraction of nodes in the largest connected component as a function of the fraction of deleted edges for the original BA network and for the corresponding optimized one when links are removed uniformly at random. The numerical simulation shows that edge rewiring, which improves robustness for high edge degree attack, does not have a negative effect on the resistance to random edge failure.

simulation is 17%, and the critical value of the fraction of removed edges at which the giant component disappears goes from about 0.425 to about 0.68. However, as soon as the mean degree increases, the protection effect disappears, giving rise to rewired networks that have about the same vulnerability as the original ones. This can be seen in Fig. 4(b) which is for $\bar{k} \approx 8$. The reason for this behavior is understandable looking at Fig. 2 above. When the graph is very sparse, rewiring edges heuristically may still increase robustness because the original robustness is not high. But for larger $\bar{k}$ the network is already intrinsically robust and rewiring edges, given that the degree distribution is kept fixed, improves robustness very little if at all, as nothing changes in a fundamental manner in the graph structure. However, as already noted, most technological real-world networks, for which robustness is a concern, are sparse with mean degrees between 2 and 5, e.g., the Internet, power grids, electronic circuits, railways, and P2P networks among others [20]. The results found in [17] using edge betweenness as a target for the attack and $N = 100$ are similar in spite of a much higher computational burden and the global information needed.

Targeted link attacks is the main subject of our study but, once rewired networks optimized against this type of perturbation have been obtained, it is of interest to find out whether the same networks can also withstand random edge failure. In fact, it is a well known result that scale-free networks are robust against this type of attack [3]. Random edge failures are very common and may represent, for instance, natural malfunctioning of power or transmission lines, a frequent phenomenon in the real world. This kind of failures may also be due to hostile actions when the attacker does not know enough about the network structure and thus strikes edges at random.

Typically, one obtains results similar to those plotted in Fig. 5 when testing the robustness of the original BA network and the corresponding optimized one against random edge removal. Clearly, there is little difference and thus the rewiring that leads to more robust graphs when submitted to high edge degree attack does not negatively affect the natural robustness of scale-free graphs when edge failures are random. In fact, the networks only fall completely apart when the fraction of edges removed is larger than 0.82.

3.2. Local modification

Unrestricted edge swapping as described in the previous section is reasonable when designing a new network or when re-routing has a low cost, as in virtual overlay networks. However, it is often unfeasible for an existing real networks such as rewiring a power line between distant nodes or establishing a new air route to an arbitrary destination because of technological and economical constraints. On the other hand, if modifications are kept local in the sense that the redirections are restricted to the neighborhood of the chosen edges, then the change could make more sense in an existing real network, assuming that graph distance is approximately representative of geographic distance, as is often the case in transportation and distribution networks for example. The swap heuristic 1 of Section 3 can be easily modified such that edges are rewired locally instead of globally and the new procedure is schematized in pseudocode 2. Fig. 6 is an example of how robustness improves after applying the algorithm to an original BA network. The robustness gain averaged over twenty runs on BA networks with $N = 200$ and $\bar{k} \approx 4$ is 16.7% i.e., about the same improvement, given the standard deviations, found for global edge swapping. This is significant as the rewiring procedure is certainly more convenient for real networks. Besides, we have checked that, as for the global rewiring case, the resulting networks are also as good as the original BA nets against random edge failure.

3.3. Using efficiency as a robustness measure

It is interesting to investigate whether optimizing BA networks with respect to another robustness measure gives results that are coherent with those found using R. This is far from clear, as can be seen from the results of the detailed

Algorithm 2 Local Edge Swap Heuristic

```
Create graph G
Compute R(G)
for i = 1 → number of iterations do
    Choose edge e_ij ∈ G uniformly at random
    Choose edge e_lk ∈ G belonging to a neighbor of i or j
                                                        ▷ Check for inconsistencies:
    if e_ij and e_lk are adjacent or new edges already exist then
        Continue with the next iteration
    else
        Swap edges e_ij, e_lk obtaining a new graph H
        Compute R(H)
    end if
    if R(H) > R(G) then
        G ← H
    end if
end for
```

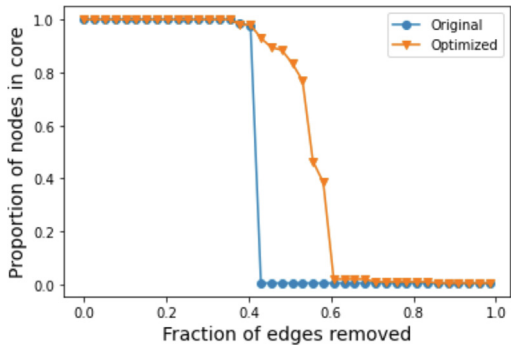

Fig. 6. Size of the largest connected component as a function of the fraction of edges removed for BA graphs with 200 nodes and mean degree $\bar{k} \approx 4$. The attack is to high degree edges as in Fig. 4 but the swapping of edges takes place locally. Comparing with Fig. 4(a) indicates that local edge swapping is as good as global swapping.

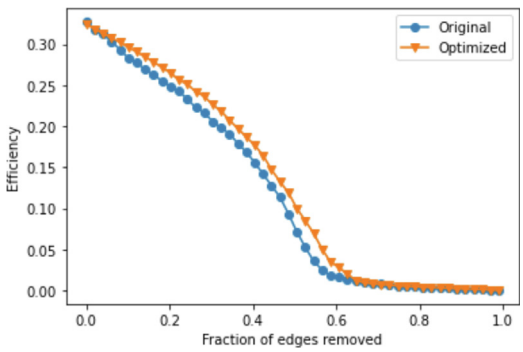

Fig. 7. (a): Network efficiency as a function of the fraction of edges removed for BA graphs of size N = 200 and mean degree $\bar{k} \approx 4$ for the original and the optimized networks. Rewired BA networks have better efficiency than the original ones in the whole range for a fraction of removed edges lower than 6.5.

study of Liu et al. [15]. Often networks optimized under one attack type or using a given robustness measure do not maintain their resiliency when submitted to another attack type or with respect to a different robustness measure. Thus we present next the results obtained using efficiency (see Section 2) as the robustness measure, and attacking high degree edges as before under the edge swap heuristic with efficiency in place of R in pseudocode 1. The results are shown in Figs. 7 and 8 for BA networks of size $N = 200$ and $\bar{k} \approx 4$. In Fig. 7 it is seen that efficiency is a smoother measure than R because, by definition, it only vanishes when all the network links have been destroyed, while R has a sudden drop as soon as the giant component disappears. The final networks obtained when targeting high degree edges are more robust

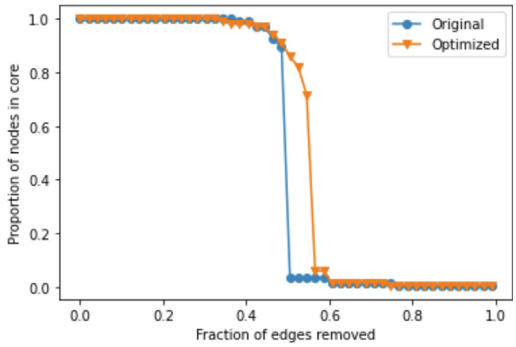

Fig. 8. Robustness of an efficiency-optimized network in terms of size of the largest connected component as a function of the fraction of removed edges. The network is slightly less robust that those originally optimized for R (see Fig. 4(a)) but still significantly better that the original network.

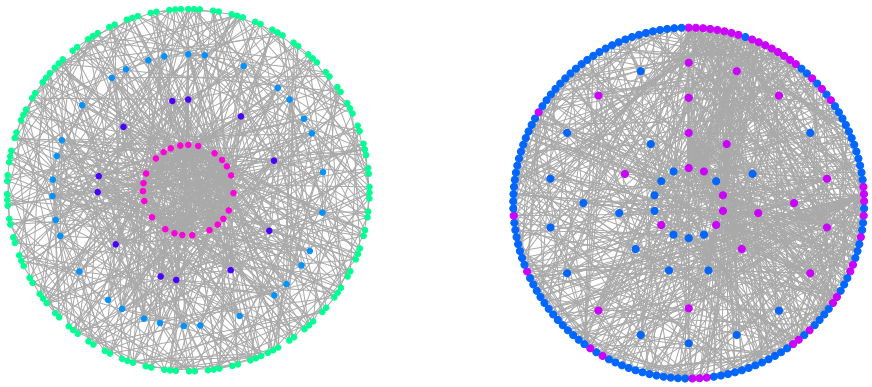

Fig. 9. Left image: an onion network obtained by the edge swap algorithm when the original BA network nodes are attacked by node degree. Right image: the structure of an optimized network of the same size and mean degree obtained when high degree edges are the attack target. Nodes with the same color have the same coreness. The network on the left image has four distinct core layers while the right network cannot be represented in the same layered fashion.

in terms of efficiency than the original ones. In addition, they are also more robust with respect to the R measure (see Fig. 8), although to a smaller extent than those directly optimized for R. This is a useful result but in practice using R instead of efficiency is preferable in computational terms as for efficiency we need to calculate all pairs shortest paths, an $O(N^2 + NM)$ algorithm [20], with N the number of nodes and M the number of links of the graph. R, on the other hand, has a cost that is dominated by the edge sort operation and is thus $O(M \ln M)$. If we take $N = 200$ and $M = (N\langle k\rangle)/2$, with $\langle k\rangle$ small (sparse graphs), setting $\langle k\rangle = 4$ we have $M \approx 2N$ and we get $3N^2$ operations to compute efficiency and $2N \ln 2N$ operations to compute R. If the algorithm performs $s \simeq 10^4$ total swap operations this gives 1.2×10^9 operations for efficiency and only $\approx 2.4 \times 10^7$ for R. In fact this is a lower bound because here we have assumed that the robustness measure is computed once and for all on the initial graph, while in many cases, including the present work, it is updated after each graph modification. In summary, different robustness measures are useful because they may expose different structural aspects of a given network but computational considerations are also important, especially for large networks.

3.4. Structure of robustness-enhanced networks

In this section we examine the main structural properties of the optimized networks with respect to the original ones. Before proceeding, it is useful to recall that using the edge swap heuristic when the highest degree node is removed first gives rise to optimized networks that have a particular structure, called "onion networks" consisting of a core of highly connected nodes, surrounded by layers of nodes of decreasing degree [9,10]. It would be nice if these networks would also be resilient with respect to edge removal but it is known that they are fragile when high betweenness edges are the target as first pointed out in [17]. For the sake of illustration, in Fig. 9 we show such an onion network together with a network of the same size and mean degree optimized with respect to attacks targeting the edge of highest degree. The difference is striking: the onion network on the left has regular layers of nodes each of which has at least k connections, with k decreasing towards the outside of the plot. Typical networks with enhanced robustness with respect to high degree link attack such as the one represented in the right image do not have this layer structure. The following Fig. 10 shows that, as found in [17] for edge betweenness, onion networks optimized to decrease vulnerability to high degree node attacks

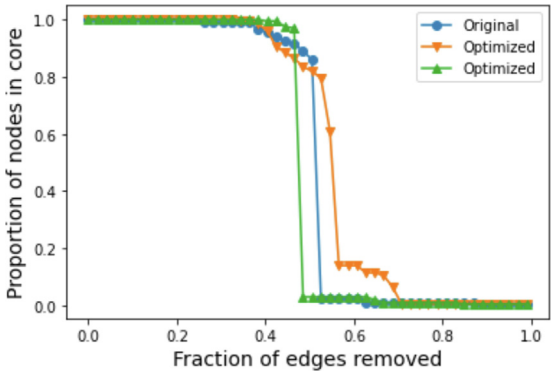

Fig. 10. Tolerance to high degree edge attacks of an original BA network (blue curve), of the network optimized to enhance robustness against high degree edge attacks (orange curve), and of the onion network resulting from optimization for high degree node attack (green curve). The two optimized networks were obtained starting from the same original BA network. The onion network which is robust against node attacks becomes the weakest one against link attacks.

Table 1
Network Properties of original BA networks and networks with enhanced robustness with respect to edge removal. From left to right: average clustering coefficient $\langle C \rangle$, diameter D, average path length $\langle L \rangle$, degree assortativity r, robustness R. The values are averages over 20 edge swap runs. Standard deviations in parentheses.

Network type	$\langle C \rangle$	D	$\langle L \rangle$	r	R
Original BA networks	0.0826 (0.018)	6.15 (0.316)	3.355 (0.085)	−0.1450 (0.036)	0.519 (0.011)
Robustness-enhanced	0.0691 (0.021)	6.75 (0.483)	3.429 (0.071)	−0.1665 (0.029)	0.621 (0.003)

are not robust against high edge degree attacks. Indeed, looking at the figure those networks (green curve) are even more vulnerable than the original BA networks (blue curve).

The converse is also partly true: networks optimized with respect to removal of edges according to edge degree are slightly more vulnerable than the original BAs against high degree node attacks. In fact, in [17] the authors suggest that networks that are more resistant against both attack types can be obtained by using the edge swap heuristic and alternating between attacks to nodes and attack to edges during the simulation. We do not pursue this interesting direction here for lack of space and we let it for future work.

Finally, Table 1 summarizes some of the main metrics that statistically characterize complex networks for the original BA graphs and the corresponding robustness-enhanced ones. The features under scrutiny are: average clustering coefficient $\langle C \rangle$, diameter D, average path length $\langle L \rangle$, assortativity coefficient r, and the robustness index R. The above statistics, except for the robustness R which increases notably, change only slightly going from BA networks to the final ones because of the constraint of invariant node degree and degree distribution function. Nevertheless, some trends are apparent. The average path length and the diameter increase slightly with respect to the original BA networks. This indicates a moderately decreasing efficiency in communication between different parts of the network. Moreover, while BA networks are already degree-disassortative by construction, the increased disassortativity of optimized networks indicates that nodes connect to others with very different degree, which also helps to limit the damage when removing a high degree link. The average clustering coefficient of BA networks is low and it decreases slightly for the optimized networks.

4. Empirical evaluation on real networks

In this section we provide an empirical evaluation by running the algorithms on some real network data. We first used the network representing the power grid of the western States of the U.S.A. The data set is from the Koblenz Network Collection KONECT [25]. A node represents a generator, a transformator, or a substation. Edges represent power supply lines. The graph is undirected, unweighted, and has 4941 nodes and 6594 edges. The mean degree is 2.669. The degree distribution approaches a power-law with exponent $\gamma = 2.246$ with a faster decaying tail. Running the global edge rewiring algorithm 1 with 2000 rewires we obtained the results depicted in Fig. 11

Because the mean degree is small, the critical fraction f of removed edges, i.e., the f at which the giant connected component disappears is low. However, the gain in robustness is substantial: R improves notably and f goes from about 0.23 to about 0.42, as can be seen in the figure. There is little change in the network statistics: the degree distribution function stays the same by definition and there is very little degree correlation, with a low clustering coefficient, as in the original network. The diameter and the average path length also remain very similar to the original values. If one limits the number of rewired edges to 1000 the results are already appreciable: the critical connectivity point goes from 0.23 to about 0.36. Using more than 3000 rewires gives decreasing returns in robustness and is not useful. Finally, if we use the

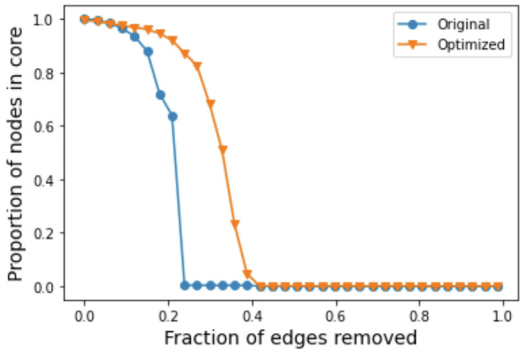

Fig. 11. Relative size of the largest connected component as a function of the fraction of removed edges when edges are removed highest edge degree first for the undirected network representing the power grid of the western States of the U.S.A. The network has 4941 vertices and 6594 edges. The degree distribution is approximated by a power-law.

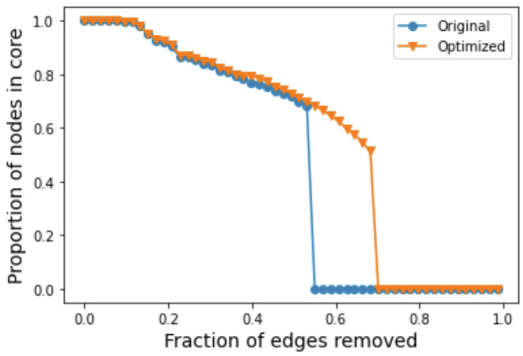

Fig. 12. Relative size of the largest connected component as a function of the fraction of removed edges for the undirected network representing a small portion of the Internet at the level of the autonomous systems (University of Oregon Route Views Project). The network has 22963 vertices and 48436 edges and the degree distribution is close to scale-free.

local rewiring algorithm 2 on the same network the results are still good but slightly worse than for the global rewiring for the same number of iterations. It would be interesting to see what is the total link length in each case but the data set does not contain geographical coordinate information for the nodes.

The next case study uses a snapshot of the Internet at the level of autonomous systems coming from the University of Oregon Route Views Project; the data set was downloaded from the Koblenz Network Collection KONECT [25]. The network is undirected and unweighted with 22963 nodes and 48436 edges for an average degree of 4.219. The degree distribution is scale-free with an exponent $\gamma = 2.435$. The network is degree-disassortative with a correlation coefficient equal to -0.19838, has a diameter of 11, and a mean average path length of 3.738.

Fig. 12 shows that the rewired network is way more resistant to targeted edge attacks than the original one with only 2000 edges rewired, i.e., less than 10%. The other characteristics of the rewired network do not change significantly with respect to the original one.

5. Conclusions

Network resilience against targeted attacks directed to network edges are important for network integrity but they have been less studied than attacks to vertices, probably because removal of vertices affects a network in a more radical manner than deliberate damage inflicted to links. However, we have argued that links can be an important target; for example, cutting a power line is easier than destroying a power plant. Thus the problem of enhancing the robustness of a network against such malicious attacks is also very important. Here we worked under the assumption that the individual node degrees, and thus the degree distribution function do not change, as in several other studies. This is a heavy constraint but it better represents the situation when dealing with existing networks and their technological and economical limitations. Several edge attack techniques and robustness measures have been described in the literature but in general they are too costly in computational terms and require global network information which may not be available. These considerations are important for, in the end, the methodology is useful only if it can be deployed in a real world situation. Therefore, our study is based on attacks and robustness measures that are both easy to compute and of local nature. For attacks we chose edges with high degree, which only requires sorting edges by degree and for robustness the

R metrics, which can be computed in linear time. We have shown that the robustness improvement on model networks obtained in this way is significant and is comparable to the results obtained with more computationally heavy approaches if the networks are sufficiently sparse. In addition, since in practice local modification is always preferable for cost reasons in existing networks, we have found that a simple modification of the basic edge swap heuristic can be made to work in the environment of a chosen edge with enhanced robustness results comparable with those corresponding to unrestricted global changes. In addition, irrespective of whether the modifications are local or on a global scale, the rewired networks are as robust as the original ones to random edge removal. We also presented two real-world network case studies: a power grid network and a portion of Internet at the level of autonomous systems. In both cases the approach was successful and produced more robust networks with the same degree distributions by rewiring only a fraction of the network edges. Finally, we note that while we worked under the strong constraint of an invariant vertex degree, another approach that admits moderate local modifications of vertices degrees, in general adding edges, has been proved useful in previous work and could be studied by methods similar to those presented here.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] A.-L. Barabási, Network Science, Cambridge University Press, Cambridge, 2016.
[2] H. Frank, I. Frisch, Analysis and design of survivable networks, IEEE Trans. Commun. Technol. 18 (5) (1970) 501–519.
[3] R. Albert, H. Jeong, A.-L. Barabási, Error and attack tolerance of complex networks, Nature 406 (2000) 378–382.
[4] M. Gosak, M. Milojević, M. Duh, K. Skok, M. Perc, Networks behind the morphology and structural design of living systems, Phys. Life Rev. 41 (2022) 1–21.
[5] M. Jusup, P. Holme, K. Kanazawa, M. Takayasu, I. Romić, Z. Wang, S. Geček, T. Lipić, B. Podobnik, L. Wang, L. Wei, T. Klanjšćek, J. Fan, S. Boccaletti, Perc. M., Social physics, Phys. Rep. 948 (2022) 1–148.
[6] R. Kumar, A. Singh, Robustness in multilayer networks under strategical and random attacks, Procedia Comput. Sci. 173 (2020) 94–103.
[7] X. Liu, E. Maiorino, A. Halu, K. Glass, R.B. Prasad, J. Loscalzo, J. Gao, A. Sharma, Robustness and lethality in multilayer biological molecular networks, Nature Commun. 11 (1) (2020) 6043.
[8] Y. Kazawa, S. Tsugawa, Effectiveness of link-addition strategies for improving the robustness of both multiplex and interdependent networks, Phys. A 545 (2020) 123586.
[9] C.M. Schneider, A.A. Moreira, J.S. Andrade Jr., S. Havlin, H.J. Herrmann, Mitigation of malicious attacks on networks, Proc. Natl. Acad. Sci. 108 (10) (2011) 3838–3841.
[10] H.J. Herrmann, C.M. Schneider, A. Moreira, J.S. Andrade, S. Havlin, Onion-like network topology enhances robustness against malicious attacks, J. Stat. Mech. Theory Exp. 2011 (01) (2011) P01027.
[11] A. Beygelzimer, G. Grinstein, R. Linsker, I. Rish, Improving network robustness by edge modification, Phys. A 357 (3–4) (2005) 593–612.
[12] P. Van Mieghem, D. Stevanović, F. Kuipers, C. Li, R. Van De Bovenkamp, D. Liu, H. Wang, Decreasing the spectral radius of a graph by link removals, Phys. Rev. E 84 (1) (2011) 016101.
[13] V.H. Louzada, F. Daolio, H.J. Herrmann, M. Tomassini, Generating robust and efficient networks under targeted attacks, in: Propagation Phenomena in Real World Networks, Springer, 2015, pp. 215–224.
[14] H. Chan, L. Akoglu, Optimizing network robustness by edge rewiring: a general framework, Data Min. Knowl. Discov. 30 (5) (2016) 1395–1425.
[15] J. Liu, M. Zhou, S. Wang, P. Liu, A comparative study of network robustness measures, Front. Comput. Sci. 11 (4) (2017) 568–584.
[16] P. Holme, B.J. Kin, C.N. Yoon, S.K. Han, Attack vulnerability of complex networks, Phys. Rev. E 65 (2002) 056109.
[17] A. Zeng, W. Liu, Enhancing network robustness against malicious attacks, Phys. Rev. E 85 (6) (2012) 066130.
[18] A.-L. Barabási, R. Albert, Emergence of scaling in random networks, Science 286 (5439) (1999) 509–512.
[19] C. Pizzuti, A. Socievole, P. Van Mieghem, Comparative network robustness evaluation of link attacks, in: International Conference on Complex Networks and their Applications, Springer, 2019, pp. 735–746.
[20] M.E.J. Newman, Networks: An Introduction, second ed., Oxford University Press, Oxford, UK, 2018.
[21] Y.-J. Deng, Y.-Q. Li, R.-R. Yin, H.-Y. Zhao, B. Liu, Efficient measurement model for critical nodes based on edge clustering coefficients and edge betweenness, Wirel. Netw. 26 (4) (2020) 2785–2795.
[22] V. Latora, M. Marchiori, Efficient behavior of small-world networks, Phys. Rev. Lett. 87 (2001) 198701.
[23] G.-S. Peng, J. Wu, Optimal network topology for structural robustness based on natural connectivity, Phys. A 443 (2016) 212–220.
[24] P. Buesser, F. Daolio, M. Tomassini, Optimizing the robustness of scale-free networks with simulated annealing, in: A. Dobnikar, U. Lotric, B. Ster (Eds.), Adaptive and Natural Computing Algorithms, volume 6594 of LCNS, Springer, Heidelberg, 2011, pp. 167–176.
[25] J. Kunegis, KONECT – The Koblenz Network Collection, in: Proc. Int. Conf. on World Wide Web Companion, 2013, pp. 1343–1350.